

光速测量

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1 2π

2π	第一次	第二次	第三次	A($10^{-5}m$)	B($10^{-5}m$)
$\Omega(MHz)$	50.407	50.407	50.407	0	577.35(Hz)
近程光程 (cm)	12.80	12.81	12.82	5.77	5.77
远程光程 (cm)	7.28	7.29	7.27	5.77	5.77
	5.18	5.19	5.20	5.77	5.77
	61.30	61.29	61.30	3.33	5.77
	3.72	3.74	3.72	6.67	5.77
	63.68	63.69	63.68	3.33	5.77
	3.70	3.72	3.72	6.67	5.77
	65.52	65.54	65.51	8.82	5.77
	3.72	3.71	3.71	3.33	5.77
	72.80	72.83	72.82	8.82	5.77
光程差 (cm)	274.1	274.19	274.11	/	/
光速 ($10^8m/s$)	2.76	2.76	2.76	/	/

我们在表格中计算出了每个分量的不确定度, 我们现在计算总的不确定度不确定度的总的计算公式为

$$U = \sqrt{(\frac{\partial f}{\partial X_1})^2 U_1^2 + (\frac{\partial f}{\partial X_2})^2 U_2^2 + \dots + (\frac{\partial f}{\partial X_n})^2 U_n^2} \quad (1)$$

我们先汇总 A 类不确定度, 我们以近程光程为例子计算

$$(\frac{\partial c}{\partial y})^2 u_{yA}^2 = (\Omega \times 2)^2 u_{yA}^2$$

则

$$u_A = \sqrt{(\frac{\partial c}{\partial y})^2 u_{A_y}^2 + (\frac{\partial c}{\partial x_1})^2 u_{A_1}^2 + (\frac{\partial f}{\partial X_2})^2 u_{A_2}^2 + \dots + (\frac{\partial c}{\partial x_9})^2 u_{A_9}^2} = 19594.71947756519m/s \quad (2)$$

$$u_B = \sqrt{(\frac{\partial c}{\partial \Omega})^2 u_{B_\Omega}^2 + (\frac{\partial c}{\partial y})^2 u_{B_y}^2 + (\frac{\partial c}{\partial x_1})^2 u_{B_1}^2 + (\frac{\partial f}{\partial X_2})^2 u_{B_2}^2 + \dots + (\frac{\partial c}{\partial x_9})^2 u_{B_9}^2} = 18406.505411701233m/s \quad (3)$$

$$u = \sqrt{u_A^2 + u_B^2} = 26884.05610906694m/s \quad (4)$$

$$c = (2.76 \times 10^8 \pm 26884.05610906694)m/s \quad (5)$$

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π	第一次	第二次	第三次	A(10^{-5})m	B(10^{-5})m
$\Omega(MHz)$	50.407	50.407	50.407	0	577.35(Hz)
近程光程 (cm)	11.10	11.10	11.11	3.33	5.77
远程光程 (cm)	7.71	7.72	7.71	3.33	5.77
	5.72	5.71	5.73	5.77	5.77
	29.34	29.35	29.33	5.77	5.77
	3.58	3.58	3.58	0	5.77
	34.21	34.22	34.20	5.77	5.77
	1.72	1.74	1.73	5.77	5.77
	26.30	26.32	26.31	5.77	5.77
	3.52	3.52	3.52	0	5.77
	29.22	29.21	29.23	5.77	5.77
光程差 (cm)	130.22	130.27	130.23	/	/
光速 ($10^8 m/s$)	2.63	2.63	2.63	/	/

$$u_A = \sqrt{\left(\frac{\partial c}{\partial y}\right)^2 u_{A_y}^2 + \left(\frac{\partial c}{\partial x_1}\right)^2 u_{A_1}^2 + \left(\frac{\partial f}{\partial X_2}\right)^2 u_{A_2}^2 + \dots + \left(\frac{\partial c}{\partial x_9}\right)^2 u_{A_9}^2} = 30056.927611154417 m/s \quad (6)$$

$$u_B = \sqrt{\left(\frac{\partial c}{\partial \Omega}\right)^2 u_{B_\Omega}^2 + \left(\frac{\partial c}{\partial y}\right)^2 u_{B_y}^2 + \left(\frac{\partial c}{\partial x_1}\right)^2 u_{B_1}^2 + \left(\frac{\partial f}{\partial X_2}\right)^2 u_{B_2}^2 + \dots + \left(\frac{\partial c}{\partial x_9}\right)^2 u_{B_9}^2} = 36812.28076852169 m/s \quad (7)$$

$$u = \sqrt{u_A^2 + u_B^2} = 47524.340214280186 m/s \quad (8)$$

$$c = (2.63 \times 10^8 \pm 47524.340214280186) m/s \quad (9)$$

3 低频仪器

2π	第一次
$\Omega(MHz)$	3.426
近程光程 (cm)	21.22
	147.60
	144.02
	143.51
	143.32
远程光程 (cm)	143.50
	139.61
	4.47
	86.51
	3.61
	94.56
光程差 (cm)	1029.49
光速 ($10^8 m/s$)	3.0577912